Geometric series

1 17

2 $\frac{5461}{64}$

3 9.143

4 683

5 $20\,475$

6

a 14.29 **b** -204.92

7

a

i $n = 10$ **ii** $S_{10} = -1364$

b

i $n = 6$ **ii** $S_6 = \frac{1365}{16}$

8

a $n = 8$ **b** $n = 9$ **c** $n = 6$ **d** $n = 6$

9

a $\frac{1}{2}$ **b** 10.0

10

a $x\left(\frac{1-\left(\frac{2}{5}x\right)^n}{1-\frac{2}{5}x}\right)$ **b** 2555

11 $\frac{1-4\times\left(\frac{1}{4}\right)^n+3\times\left(\frac{1}{5}\right)^n}{12}$

12 $r = 3$

13 $u_1 = 327\,680$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Summation notation

14

a $\sum_{k=1}^{10} 2^{k-1}$

b $\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$

c $\sum_{k=1}^{6} (-3)^k$

d $\sum_{k=1}^{5} 6\,(3)^{k-1}$

e $\sum_{k=4}^{10} \left(\frac{1}{2}\right)^k$

f $\sum_{k=1}^{7} 16\left(\frac{1}{2}\right)^k$

15

a 340

b 7205

c 31

d 4914

e -252

f 242

g $\frac{3267}{4}$

h $-\frac{40}{27}$

16

$$\sum_{k=1}^{19} 2^{k-1} x^{3k-2}$$

Technology

17 5

18 8

19

a $u_1 r^4 = 405$

b $u_1 r^5 + u_1 r^6 + u_1 r^7 = 15\,795$

c $u_1 = 5, r = 3$

d $u_n = 5 \times 3^{n-1}$

Applications

20

a $\$8191$

b $\$536\,870\,911$

21

a $\$59\,559.81$

b $\$105\,598.06$

22

a 759 nurses

b 3980 nurses

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23

a $6832\ \text{m}^3$ **b** $7168\ \text{m}^3$

24 $n = 5$

25

a $20\left(\dfrac{1-0.8^n}{0.2}\right)\ \text{m}$ **b** $59.04\ \text{m}$ **c** $9\ \text{m}$

26

a 0.04

b $5000 \times 1.04 + 700$

c $5000 \times (1.04)^2 + 700 \times 1.04 + 700$

d $5000 \times (1.04)^3 + 700 \times (1.04)^2 + 700 \times 1.04 + 700$

e $5000 \times (1.04)^n + \dfrac{700\left((1.04)^n - 1\right)}{1.04 - 1}$

f $\$13\,293$

27

a

i $50\,000 \times 1.059 + 3000$

ii $50\,000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

iii $50\,000 \times (1.059)^3 + 3000 \times (1.059)^2 + 3000 \times 1.059 + 3000$

b $\dfrac{3000\left((1.059)^n - 1\right)}{0.059}$

c $\$99\,791$

28

a $A_1 = 30\,000 \times 1.015 - 600$

b

i $A_2 = 30\,000 \times (1.015)^2 - (1.015 \times 600) - 600$

ii $A_3 = 30\,000 \times (1.015)^3 - \left(1.015^2 \times 600\right) - (1.015 \times 600) - 600$

c $A_n = 30\,000 \times (1.015)^n - \dfrac{600\left((1.015)^n - 1\right)}{1.015 - 1}$

d $\$22\,908.60$

29

a $1400 \times (1.005)^n + \dfrac{165\left((1.005)^n - 1\right)}{(1.005) - 1}$

b $\$8165.81$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

30

a $A_1 = 190\,000 \times 1.07 - x$

b

i $A_2 = 190\,000 \times (1.07)^2 - 1.07x - x$

ii $A_3 = 190\,000 \times (1.07)^3 - (1.07)^2\, x - 1.07x - x$

c $190\,000 \times (1.07)^n - \dfrac{x\left((1.07)^n - 1\right)}{0.07}$

d $\$17\,934.66$

31

a $110\,000 \times 1.04^n - \dfrac{x\left(1.04^n - 1\right)}{0.04}$

b $x = 4488.88$

32

a $190\,000 \times 1.0025^n - \dfrac{x\left(1.0025^n - 1\right)}{0.0025}$

b $x = 801.05$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647